

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CSG

NEWTON'S LAWS OF MOTION

LECTURE-3

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1 Revision ✓
- 2 Friction ✓
- 3 Type of Friction ✓
- 4 Dynamics of Uniform Circular Motion ✓ (R)
- 5 Centripetal Force ✓

I'm the best ✓

I can do it alone ✓

God is always with me ✓

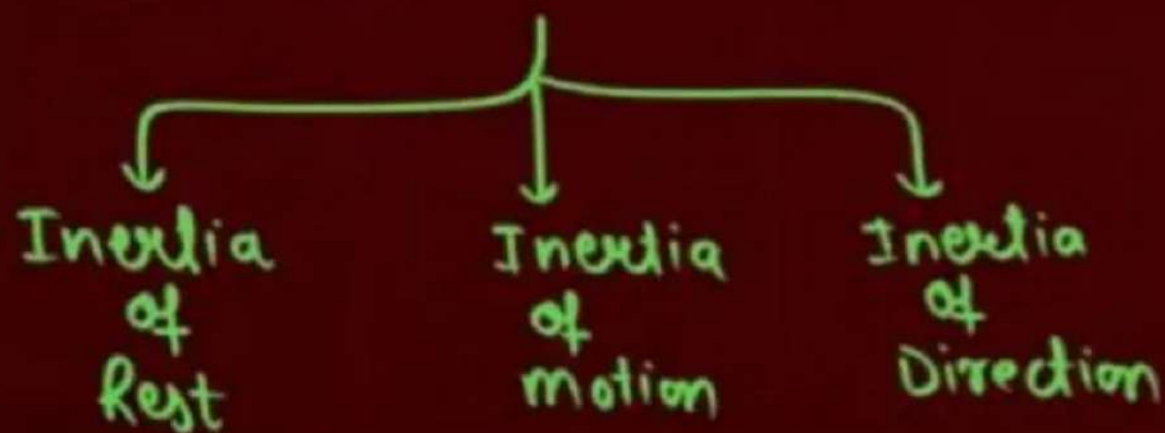
I am a winner ✓

Today is my day ✓

Recap From 1st Lecture

Inertia (1st law of motion)

Property of Body by virtue of which Body cannot change its state of Rest, motion and Direction



Momentum (p)

$$\vec{p} = m\vec{v} \quad \text{kg m/s vector}$$

$$\bullet p_f = mv \quad \bullet p_i = mv$$

② # Newton 2nd laws of Motion

$$F = \frac{\Delta p}{\Delta t} = ma \rightarrow \text{qth}$$

$$F = \frac{\Delta p}{\Delta t} = \frac{dp}{dt} = m \frac{dv}{dt} + v \frac{dm}{dt}$$

③ Impulse

Force acting for the short period of time.

$$\uparrow F = \frac{\Delta p}{\Delta t} \downarrow$$

$$F = \frac{dp}{dt}$$

$$\boxed{F dt} = dp$$

Impulse

Impulse momentum theorem

Recap From 2nd Lecture

Newton third law of motion

Action = -Reaction

- * Rocket propulsion
- * walking
- * Boat
- * Recoiling on gun
- * Flying of Bird.

law of conservation of momentum

if $F_{ext} = 0$

$$P_i = P_f$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Application of law of conservation of momentum

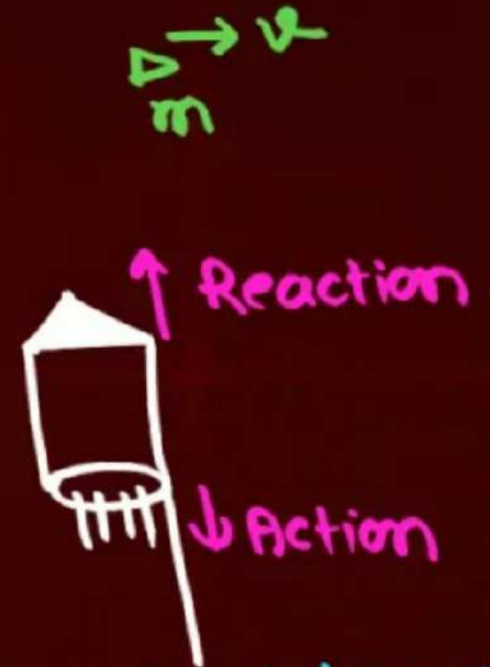
① Recoiling velocity of gun

$$V = -\frac{mv}{M}$$



② Rocket Propulsion

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$



? why Newton 2nd law is Mas Real law of Motion

$$F = \frac{dp}{dt}$$

2nd law derive $V = u$

3rd law Derived
 $f_{12} = -f_{21}$

Type of Force

① Gravitational Force (9th)

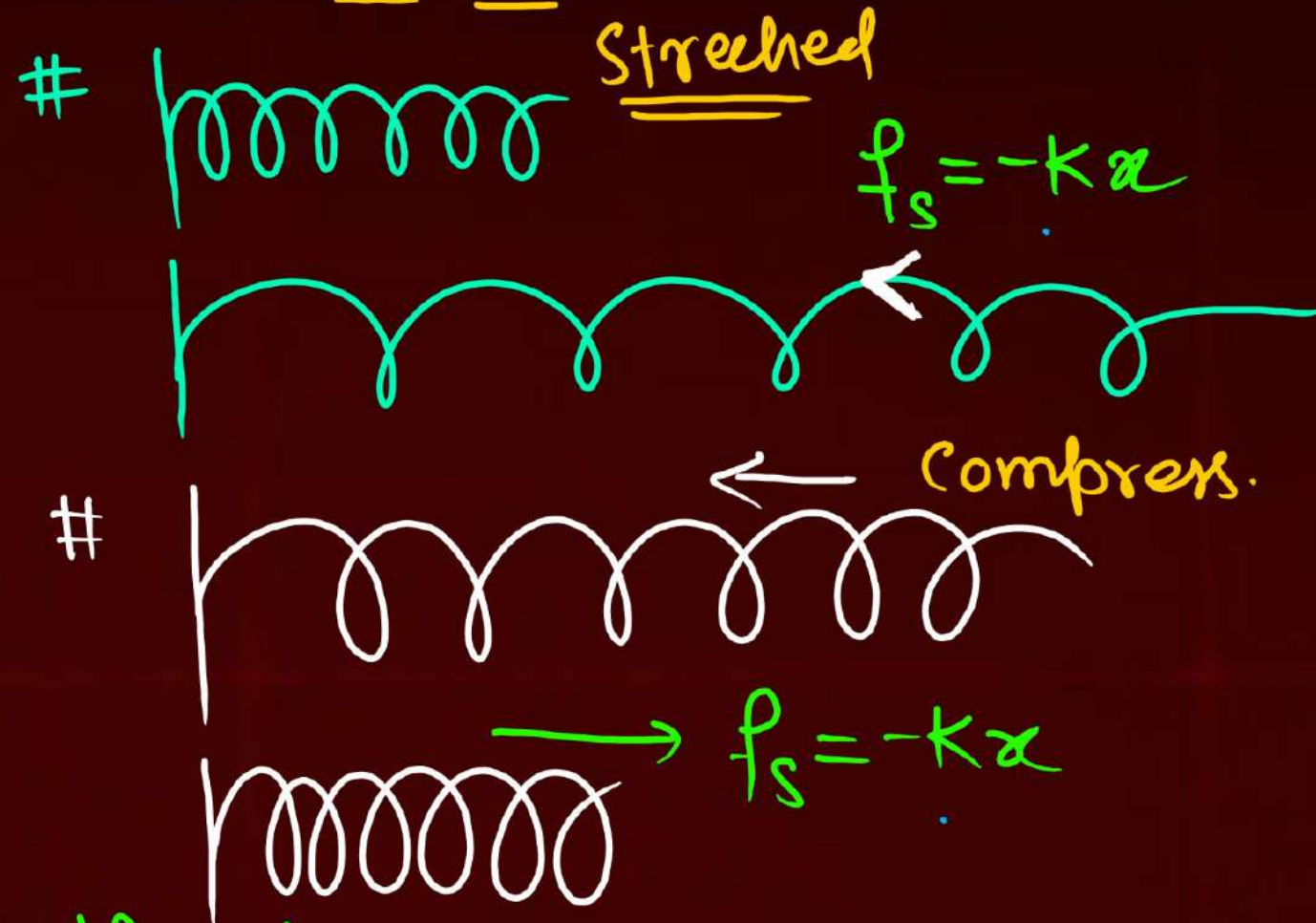
Every mass attracts
Every other mass.



$$F = G \frac{Mm}{r^2} \text{ (11th)}$$

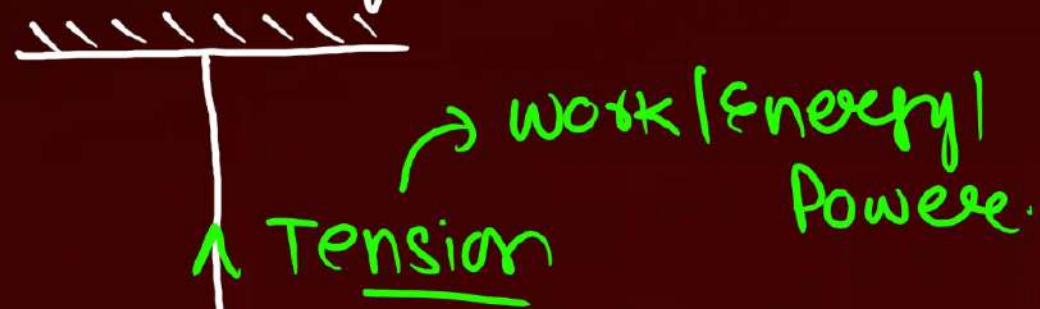
② Spring force (Work | Energy | Power)

The force developed in a
Spring when it is
Stretched or compressed.



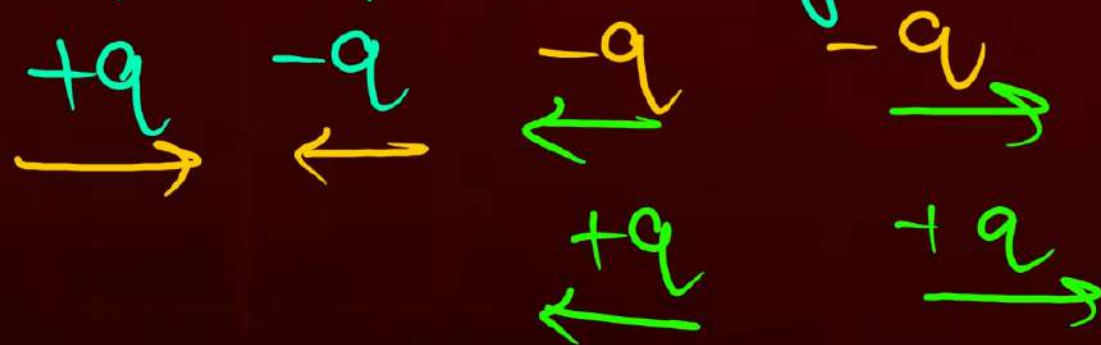
③ Tension

The force which developed in String (Rassi)



④ Electrostatic force

The force b/w the charges.



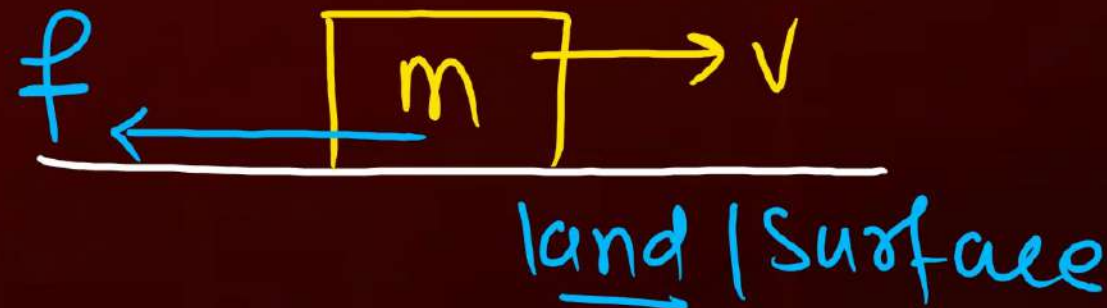
⑤ Normal Force. (contact force)

The force which come into play when one object touch the other object



⑥ Friction force (8th)

The force which oppose the Relative motion.



Friction

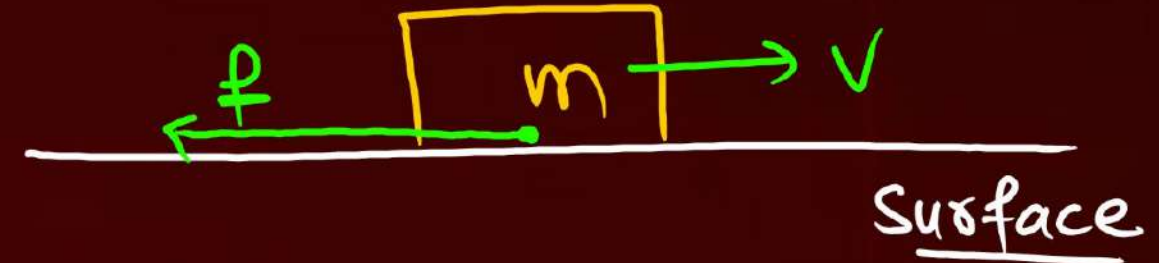
2021 2019, 2018
2025 - 5 mark

#



Friction is a contact force that opposes the relative motion or the tendency of motion between two surfaces in contact.

- ✓ Acts along the surface.
- ✓ Opposite to direction of motion or applied force.
- ✓ Always tries to resist movement.



Causes of Friction

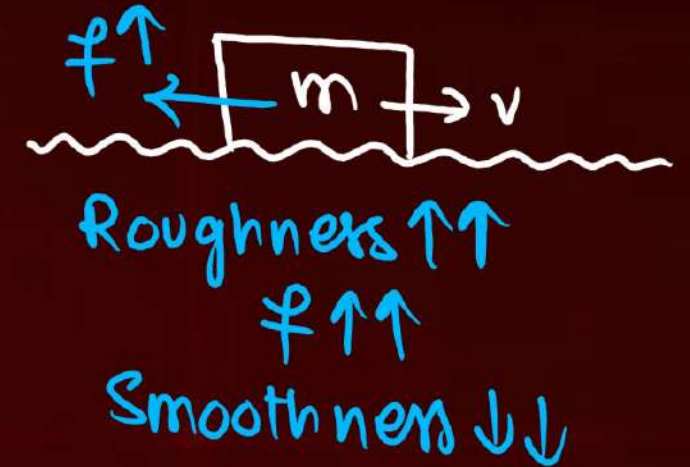
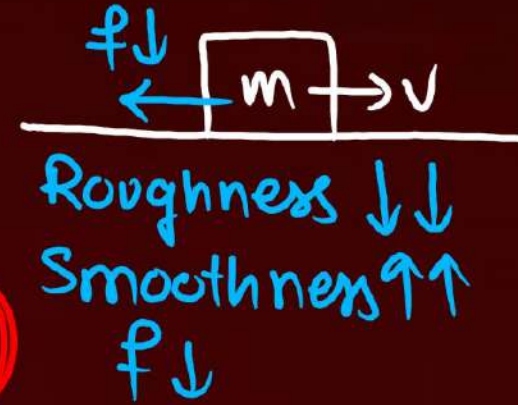
1. Surface Roughness $R \uparrow \Rightarrow f \uparrow$

Even smooth surfaces have microscopic irregularities.

2. Adhesion between molecules

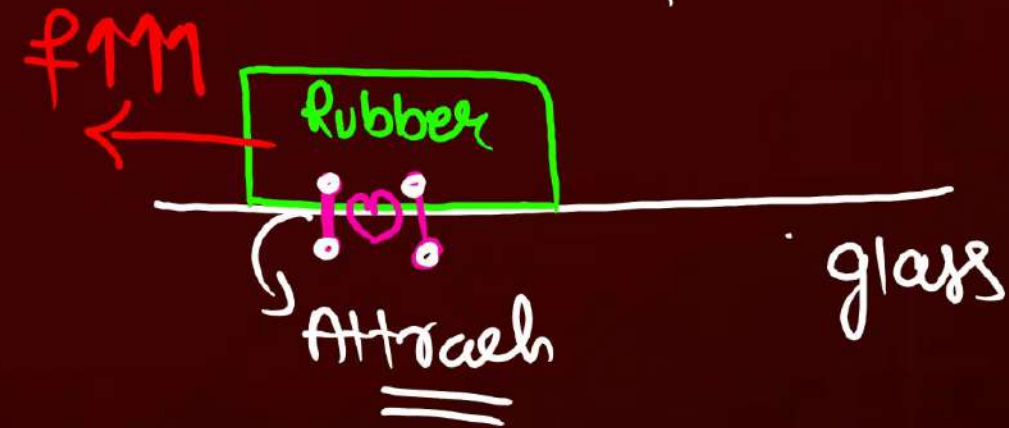
Attraction between molecules of surfaces.

① Roughness



② Adhesion b/w the molecules

Extra Smoothness

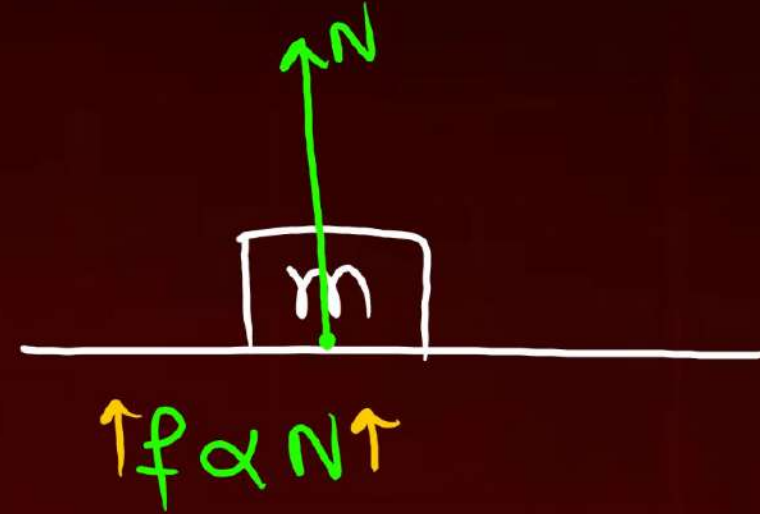


Laws Of Friction

2023 2019
2024 2016 #



1. ✓ Friction is proportional to normal reaction (force) (N)
2. ✓ Friction is independent of contact area (A^X)
 do not
3. ✓ Friction depends on nature of surfaces (μ)



$$\# f \propto N$$

$$\uparrow f = \uparrow \mu N \quad \# \text{ nature of surface}$$

μ → coefficient of friction.

tell us how much
Surface is Rough.

$R \uparrow \mu \uparrow f \uparrow$

$$f = \mu N$$

$$f \propto N$$

$$f = \ell N$$



(Numerical)

Q what are the dimensions of (a) coefficient of friction?

(Neet - 2016)

Ans $\overset{\curvearrowleft}{f} = \mu \overset{\curvearrowright}{N}$

$$\frac{f}{N} = \mu$$

$$\frac{\cancel{MLT^{-2}}}{\cancel{MLT^{-2}}} = \mu$$

$$1 = \mu \quad (\text{Dimensionless})$$

Types of Friction

- ① Static friction (deep)
- ② kinetic friction
- ③ Rolling friction.

Static Friction

- Acts when body is at rest
- Prevents motion from starting
- Adjust itself according to applied force

→ Intelligent force (mca) 2022

→ The Maximum value of Static friction is blan
limiting friction $f_s = 11.5 \text{ N}$

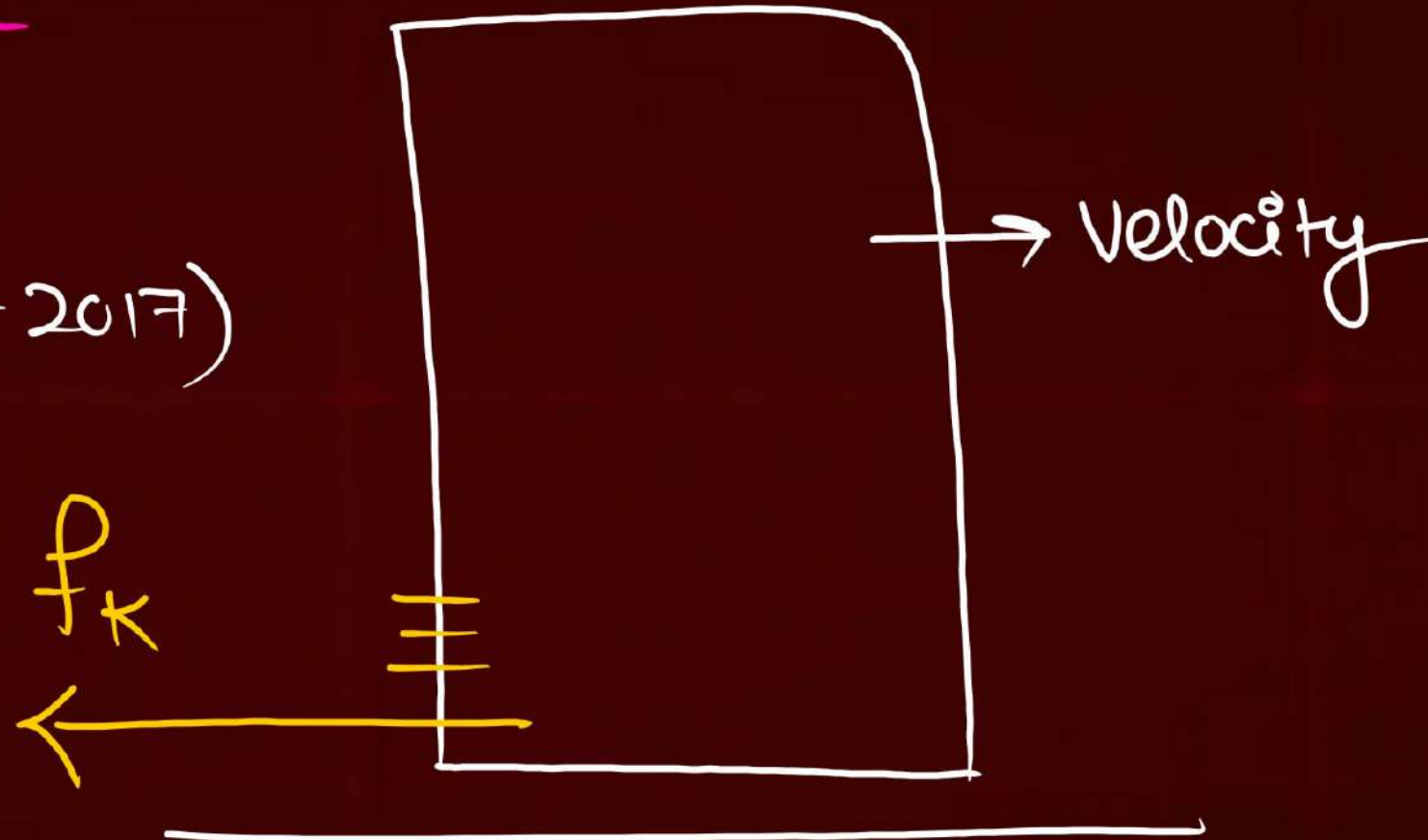


② Kinetic Friction

- Acts when body is in motion
- Constant in magnitude

$$f_k < f_s \text{ (Neet - 2017)}$$

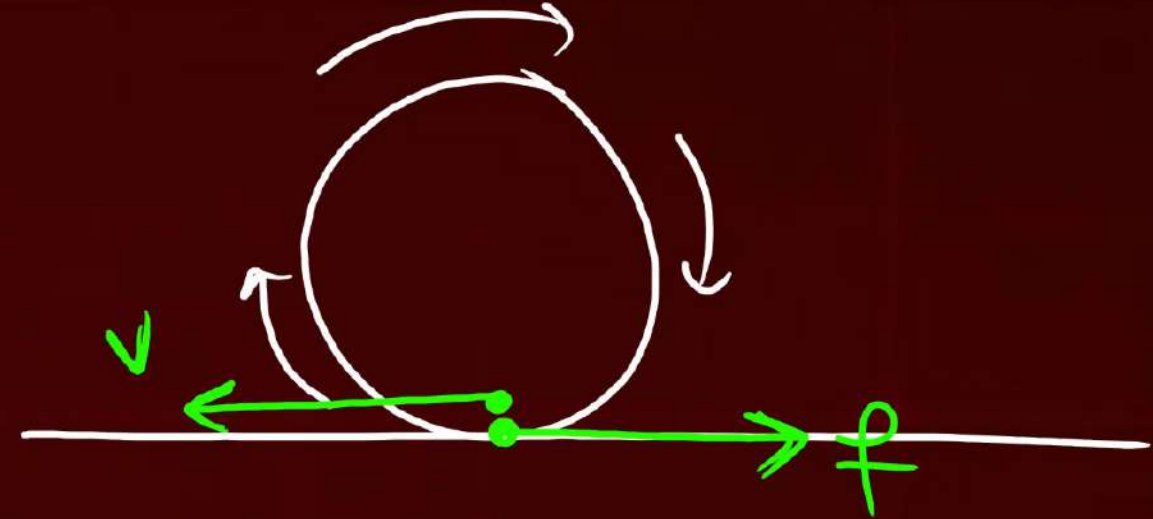
$f_k = \mu_k N$



③ Rolling Friction

- Occurs when a body rolls over a surface
- Much smaller than sliding friction (Rotation)

$$f_r = \mu_r N$$

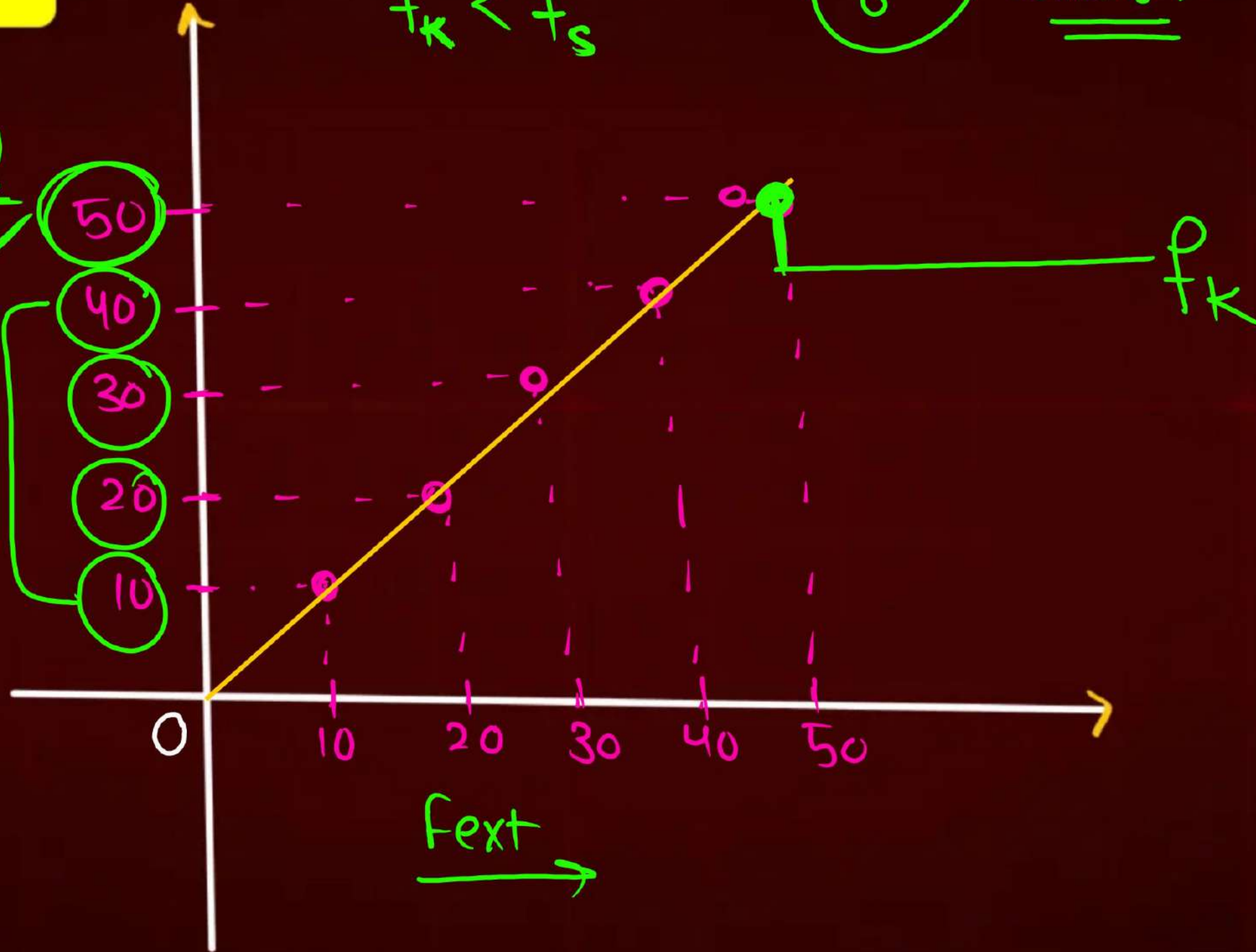


Graph

8th

✓
 $\frac{p}{f_s}$
✓

friction
 f_s



Advantages of Friction

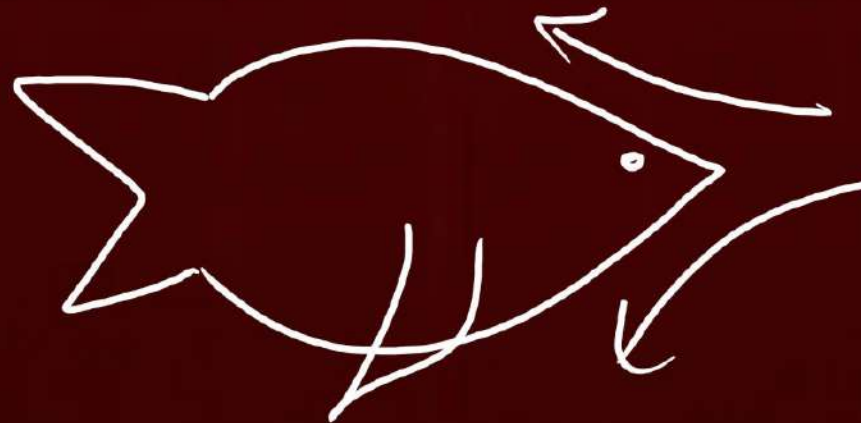
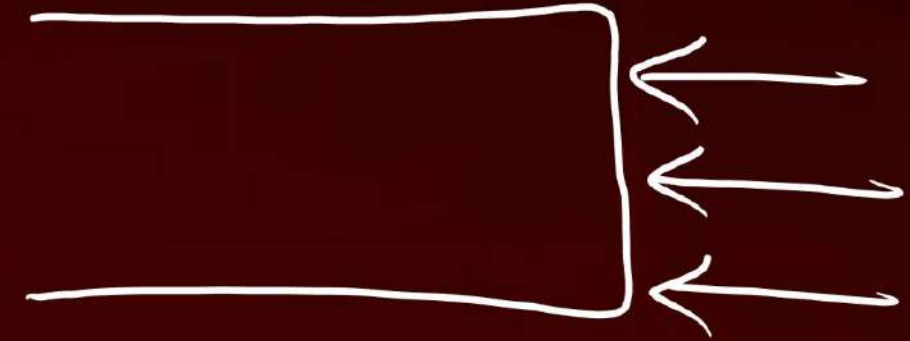
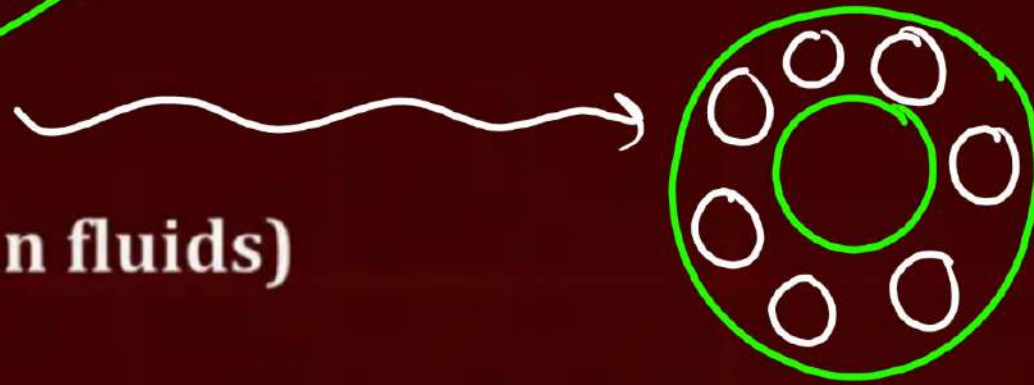
- Helps in walking ✓
- Enables writing ✓
- Vehicles move due to friction ✓
- Helps in gripping objects ✓
- Brakes work due to friction ✓

Disadvantages Of Friction

- Causes wear and tear of parts ✓ of machine
- Produces heat → energy loss ✓
- Reduces efficiency of machines ✓

Methods Of Reducing Friction

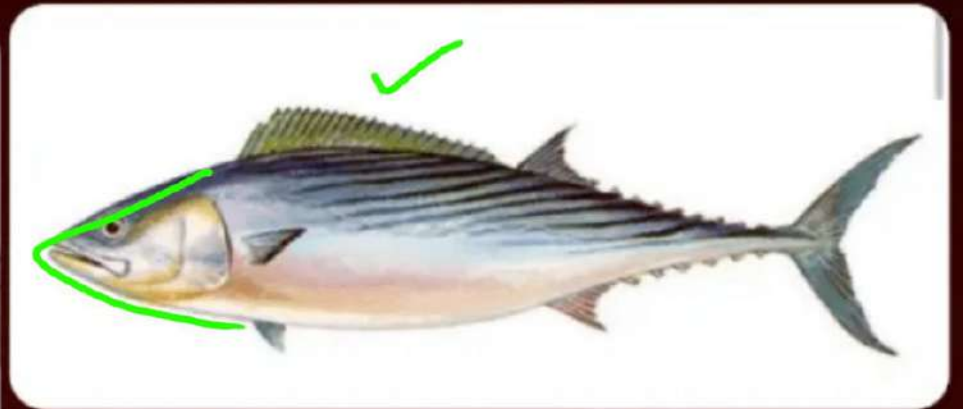
- Lubrication (oil, grease)
- Polishing surfaces
- Ball bearings
- Streamlining (in fluids)



Ball Bearings



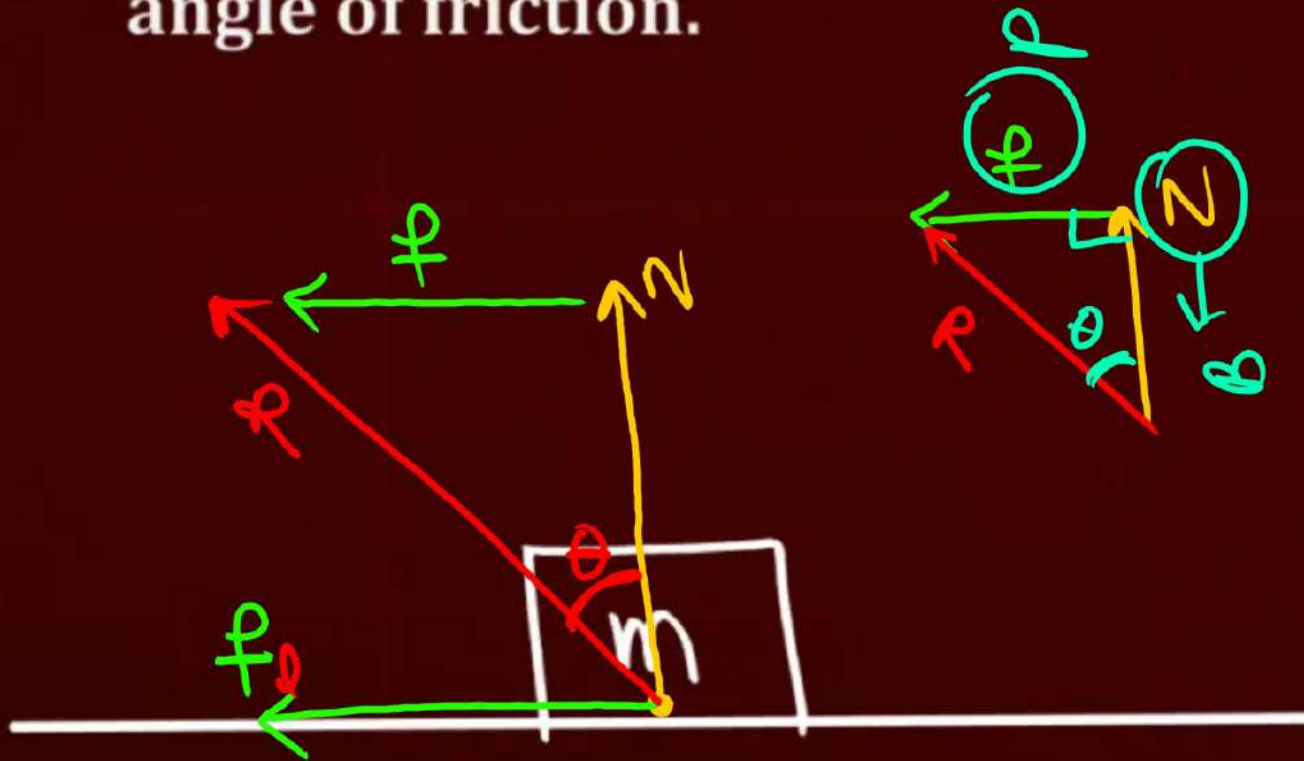
Streamlining



Angle of Friction

2 mark
(2018)

- The angle which the resultant of limiting friction and normal makes with the normal reaction is called angle of friction.



$$\tan \theta = \frac{P}{B}$$

$$\tan \theta = \frac{f}{N}$$

$$\tan \theta = \frac{\mu N}{N}$$

$$\tan \theta = \mu$$

Angle of friction

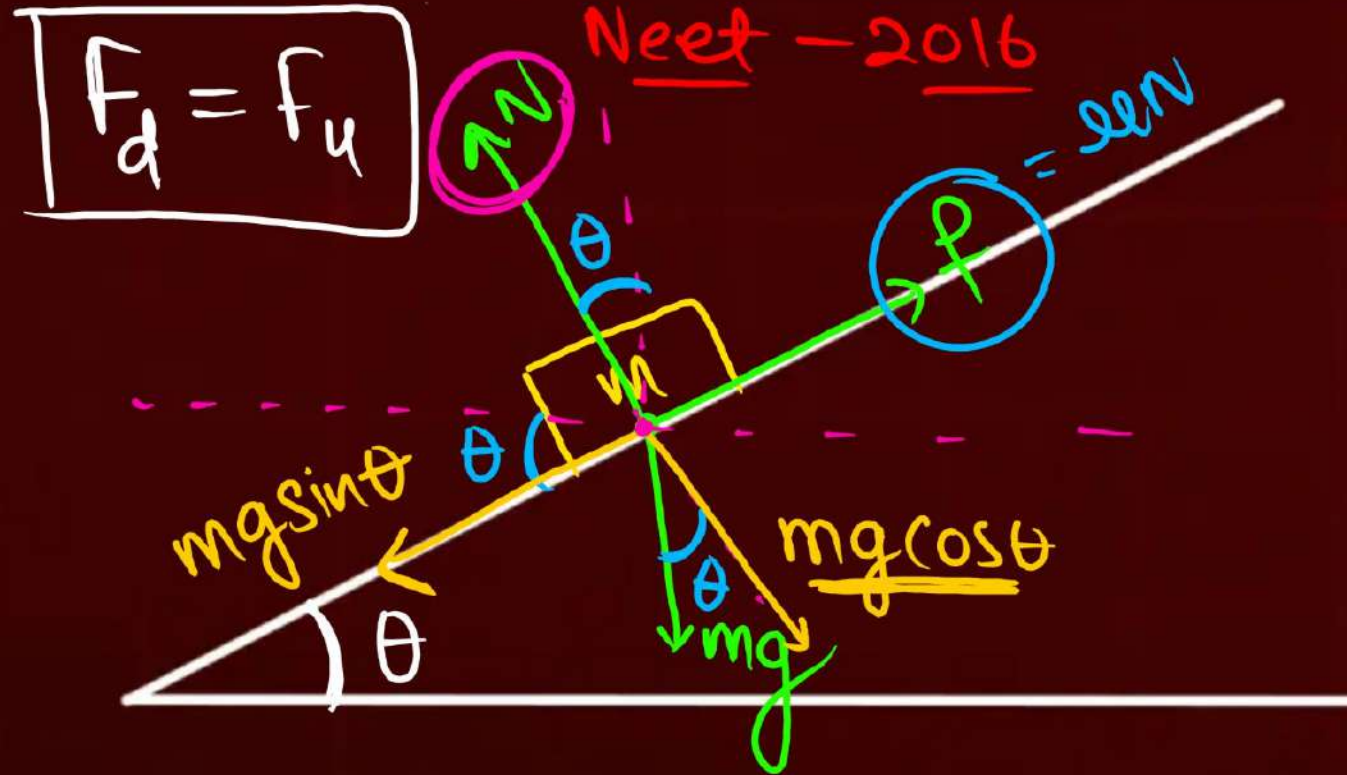
$$f = \mu N$$

Angle of Repose

(2022)

→ 2 mark.

- The minimum angle at which a body just starts sliding on an inclined plane is called Angle of repose.



$$\# F_u = F_d$$

$$\mu N = mg \sin \theta$$

$$N = mg \cos \theta$$

$$\cancel{\mu} mg \cos \theta = \cancel{mg} \sin \theta$$

$$\mu = \frac{\sin \theta}{\cos \theta}$$

$$\mu = \tan \theta$$

Most Important. (Many derivation have this concept) (Neet & Jee)



$$\theta + \alpha = 90^\circ$$

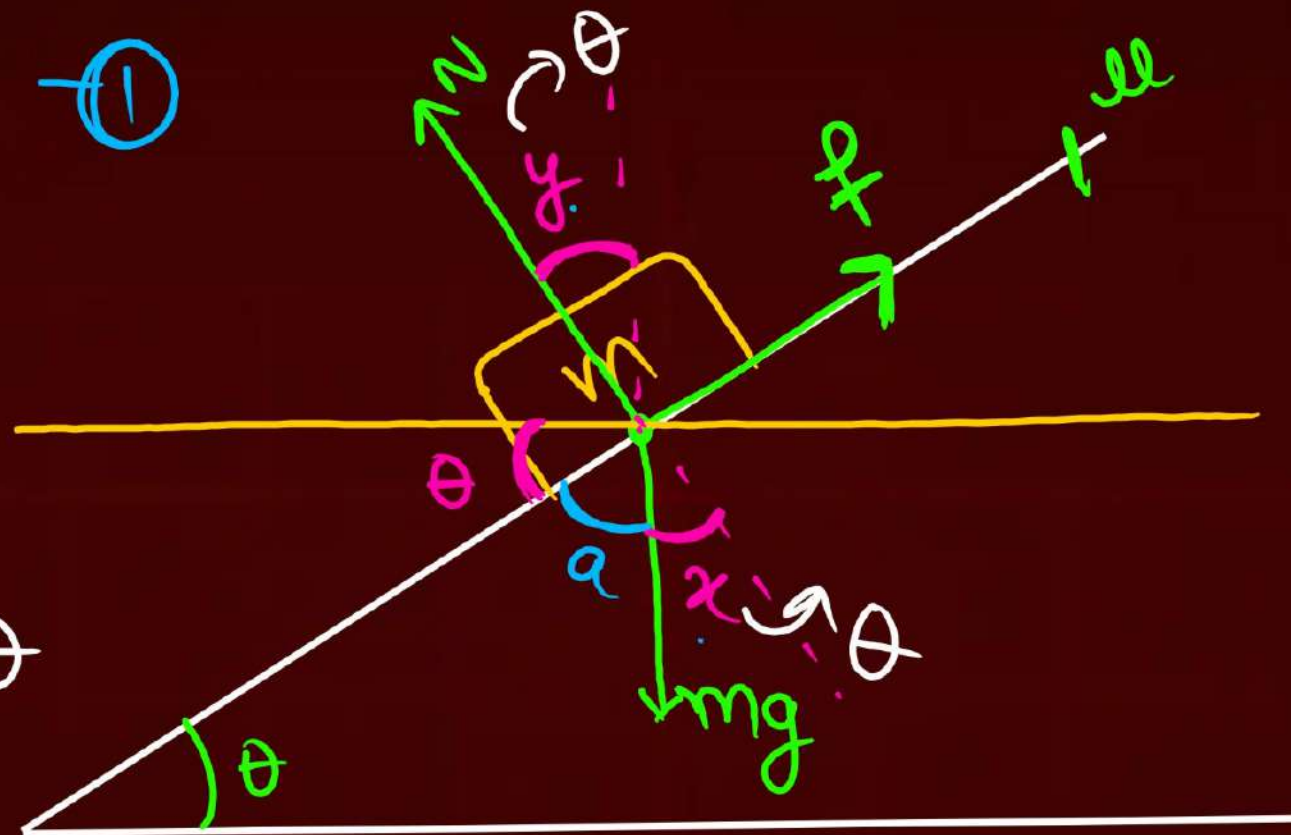
$$\alpha = 90^\circ - \theta \quad \text{--- (1)}$$

$$\underline{\alpha + \chi = 90}$$

$$90 - \theta + \chi = 90$$

$$\chi = 90 - 90 + \theta$$

$$\boxed{\chi = \theta}$$



Chocolate Question

(TKBOSE - 2018)

A block is placed on a rough horizontal surface. A force is applied, but the block does not move. Which of the following is correct?



- A** Friction is zero
- B** Friction is equal to the applied force
- C** Friction is equal to μN
- D** Friction is greater than the applied force

Thank
You